THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET 1

Chapter 11: **Areas Related to Circles** | Class 10 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** ₹ 880	Cost $= 88 \times 10$; Cost $=$ ₹ 880
2	**(c)** 50	Distance 66 m $= 6600$ cm; revolutions $= \frac{6600}{132}$; Revolutions $= 50$
3	**(b)** 28 cm	$R = r_1 + r_2 = 19 + 9$; $R = 28$ cm
4	**(a)** 10 cm	$R^2 = 6^2 + 8^2 = 100$; $R = 10$ cm
5	**(a)** 115.5 cm²	Area $= \frac{30}{360} \times \frac{22}{7} \times 21^2$; Area $= 115.5$ cm^2
6	**(b)** 38.5 m²	Area $= \frac{90}{360} \times \frac{22}{7} \times 7^2$; Area $= 38.5$ m^2
7	**(a)** 60°	$\theta = \frac{231 \times 360}{\frac{22}{7} \times 21^2}$; $\theta = 60^\circ$
8	**(d)** ₹ 660	Cost $= 132 \times 5$; Cost $=$ ₹ 660
9	**(a)** 50	Distance 110 m $= 11000$ cm; revolutions $= \frac{11000}{220}$; Revolutions $= 50$
10	**(a)** 14 cm	$R = r_1 + r_2 = 8 + 6$; $R = 14$ cm

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Angle turned in 15 minutes $= 90^\circ$; Arc $= \frac{90}{360} \times 2 \times \frac{22}{7} \times 14$ (1); Distance $= 22$ cm (1).	1+1
12	Perimeter of the square $=$ circumference $= 2\pi r = 88$ cm; Side $= 88 \div 4 = 22$ cm (1); Area $= 22^2 = 484$ cm^2 (1).	1+1
13	Grazed area is a quadrant: $\frac{1}{4}\pi r^2$; Increase $= \frac{1}{4} \times \frac{22}{7}(14^2 - 7^2)$ (1); Increase $= 115.5$ m^2 (1).	1+1
14	Angle turned in 30 minutes $= 180^\circ$; Arc $= \frac{180}{360} \times 2 \times \frac{22}{7} \times 14$ (1); Distance $= 44$ cm (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Segment $=$ sector $-$ triangle (1); Sector $= 38.5$ cm^2, triangle $= \frac{1}{2} \times 7 \times 7 = 24.5$ cm^2 (1); Segment $= 38.5 - 24.5 = 14$ cm^2 (1).	1+1+1
16	Perimeter $=$ arc $+ 2r$ (1); Arc $= \frac{90}{360} \times 2\pi \times 14 = 22$ cm (1); Perimeter $= 22 + 28 = 50$ cm (1).	1+1+1
17	The 8 ribs divide the circle into 8 equal sectors (1); Area of the circle $= \frac{22}{7} \times 28^2 = 2464$ cm^2 (1); Area between ribs $= 2464 \div 8 = 308$ cm^2 (1).	1+1+1
	OR: Segment $=$ sector $-$ triangle (1); Sector $= 154$ cm^2, triangle $= \frac{1}{2} \times 14 \times 14 = 98$ cm^2 (1); Segment $= 154 - 98 = 56$ cm^2 (1).	1+1+1
18	Perimeter $=$ arc $+ 2r$ (1); Arc $= \frac{180}{360} \times 2\pi \times 14 = 44$ cm (1); Perimeter $= 44 + 28 = 72$ cm (1).	1+1+1
	OR: The 6 ribs divide the circle into 6 equal sectors (1); Area of the circle $= \frac{22}{7} \times 21^2 = 1386$ cm^2 (1); Area between ribs $= 1386 \div 6 = 231$ cm^2 (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Segment $=$ sector $-$ triangle (2); Sector $= 102.67$ cm^2; triangle $= \frac{\sqrt{3}}{4}r^2 = 84.77$ cm^2 (2); Segment $= 102.67 - 84.77 = 17.90$ cm^2 (1).	2+2+1
	OR: Wire $=$ circumference $+$ the diameters (2); Circumference $= \pi d = 110$ mm; diameters $= 5 \times 35 = 175$ mm (2); Total $= 285$ mm (1).	2+2+1
20	Segment $=$ sector $-$ triangle (2); Sector $= 231.00$ cm^2; triangle $= \frac{\sqrt{3}}{4}r^2 = 190.73$ cm^2 (2); Segment $= 231.00 - 190.73 = 40.27$ cm^2 (1).	2+2+1
	OR: Wire $=$ circumference $+$ the diameters (2); Circumference $= \pi d = 88$ mm; diameters $= 4 \times 28 = 112$ mm (2); Total $= 200$ mm (1).	2+2+1

Mistakes to avoid:

- Using πr^2 in the arc-length formula or $2\pi r$ in the sector-area formula.
- Forgetting to add the two radii when asked for the *perimeter* of a sector.
- Using diameter in place of radius in πr^2.

- Subtracting the wrong figure for a segment: minor segment $=$ sector $-$ triangle, not sector $+$ triangle.
- Mixing units (cm and m) or writing cm instead of cm^2 for areas.
- Thinking the area of a circle doubles when the radius doubles; it becomes four times.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

MASTERMIND

THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET 2

Chapter 11: **Areas Related to Circles** | Class 10 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(d)** 13 cm	$R^2 = 5^2 + 12^2 = 169$; $R = 13$ cm
2	**(d)** 231 cm²	Area $= \frac{60}{360} \times \frac{22}{7} \times 21^2$; Area $= 231$ cm^2
3	**(a)** 154 m²	Area $= \frac{90}{360} \times \frac{22}{7} \times 14^2$; Area $= 154$ m^2
4	**(c)** 90°	$\theta = \frac{346.5 \times 360}{\frac{22}{7} \times 21^2}$; $\theta = 90^\circ$
5	**(d)** ₹ 880	Cost $= 220 \times 4$; Cost $=$ ₹ 880
6	**(d)** 50	Distance 44 m $= 4400$ cm; revolutions $= \frac{4400}{88}$; Revolutions $= 50$
7	**(b)** 17 cm	$R = r_1 + r_2 = 12 + 5$; $R = 17$ cm
8	**(b)** 15 cm	$R^2 = 9^2 + 12^2 = 225$; $R = 15$ cm
9	**(b)** 346.5 cm²	Area $= \frac{90}{360} \times \frac{22}{7} \times 21^2$; Area $= 346.5$ cm^2
10	**(a)** 346.5 m²	Area $= \frac{90}{360} \times \frac{22}{7} \times 21^2$; Area $= 346.5$ m^2

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Perimeter of the square = circumference $= 2\pi r = 176$ cm; Side $= 176 \div 4 = 44$ cm (1); Area $= 44^2 = 1936$ cm^2 (1).	1+1
12	Grazed area is a quadrant: $\frac{1}{4}\pi r^2$; Increase $= \frac{1}{4} \times \frac{22}{7}(21^2 - 14^2)$ (1); Increase $= 192.5$ m^2 (1).	1+1
13	Angle turned in 45 minutes $= 270^\circ$; Arc $= \frac{270}{360} \times 2 \times \frac{22}{7} \times 14$ (1); Distance $= 66$ cm (1).	1+1
14	Perimeter of the square = circumference $= 2\pi r = 220$ cm; Side $= 220 \div 4 = 55$ cm (1); Area $= 55^2 = 3025$ cm^2 (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Segment = sector − triangle (1); Sector $= 346.5$ cm^2, triangle $= \frac{1}{2} \times 21 \times 21 = 220.5$ cm^2 (1); Segment $= 346.5 - 220.5 = 126$ cm^2 (1).	1+1+1
16	Perimeter = arc $+ 2r$ (1); Arc $= \frac{45}{360} \times 2\pi \times 14 = 11$ cm (1); Perimeter $= 11 + 28 = 39$ cm (1).	1+1+1
17	The 10 ribs divide the circle into 10 equal sectors (1); Area of the circle $= \frac{22}{7} \times 35^2 = 3850$ cm^2 (1); Area between ribs $= 3850 \div 10 = 385$ cm^2 (1).	1+1+1
	OR: Segment = sector − triangle (1); Sector $= 616$ cm^2, triangle $= \frac{1}{2} \times 28 \times 28 = 392$ cm^2 (1); Segment $= 616 - 392 = 224$ cm^2 (1).	1+1+1
18	Perimeter = arc $+ 2r$ (1); Arc $= \frac{270}{360} \times 2\pi \times 14 = 66$ cm (1); Perimeter $= 66 + 28 = 94$ cm (1).	1+1+1
	OR: The 12 ribs divide the circle into 12 equal sectors (1); Area of the circle $= \frac{22}{7} \times 42^2 = 5544$ cm^2 (1); Area between ribs $= 5544 \div 12 = 462$ cm^2 (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Segment = sector − triangle (2); Sector $= 410.67$ cm^2; triangle $= \frac{\sqrt{3}}{4} r^2 = 339.08$ cm^2 (2); Segment $= 410.67 - 339.08 = 71.59$ cm^2 (1).	2+2+1
	OR: Wire = circumference + the diameters (2); Circumference $= \pi d = 132$ mm; diameters $= 6 \times 42 = 252$ mm (2); Total $= 384$ mm (1).	2+2+1
20	Segment = sector − triangle (2); Sector $= 205.33$ cm^2; triangle with sides r, r and angle 120° has area $\frac{1}{2} r^2 \sin 120^\circ = \frac{\sqrt{3}}{4} r^2 = 84.77$ cm^2 (2); Segment $= 205.33 - 84.77 = 120.56$ cm^2 (1).	2+2+1
	OR: The diagonal of the square is the diameter $2r$, so its area is $\frac{(2r)^2}{2} = 2r^2$ (2); Circle $= 154$ cm^2; square $= 98$ cm^2 (2); Difference $= 56$ cm^2 (1).	2+2+1

Mistakes to avoid:

- Using πr^2 in the arc-length formula or $2\pi r$ in the sector-area formula.
- Forgetting to add the two radii when asked for the *perimeter* of a sector.
- Using diameter in place of radius in πr^2.

- Subtracting the wrong figure for a segment: minor segment $=$ sector $-$ triangle, not sector $+$ triangle.
- Mixing units (cm and m) or writing cm instead of cm^2 for areas.
- Thinking the area of a circle doubles when the radius doubles; it becomes four times.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

MASTERMIND

THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET 3

Chapter 11: **Areas Related to Circles** | Class 10 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(c)** 30°	$\theta = \frac{115.5\times360}{\frac{22}{7}\times21^2}$; $\theta = 30^\circ$
2	**(a)** ₹ 2640	Cost $= 176 \times 15$; Cost = ₹ 2640
3	**(c)** 50	Distance 88 m $= 8800$ cm; revolutions $= \frac{8800}{176}$; Revolutions $= 50$
4	**(d)** 22 cm	$R = r_1 + r_2 = 15 + 7$; $R = 22$ cm
5	**(b)** 25 cm	$R^2 = 7^2 + 24^2 = 625$; $R = 25$ cm
6	**(c)** 462 cm²	Area $= \frac{120}{360} \times \frac{22}{7} \times 21^2$; Area $= 462$ cm^2
7	**(a)** 616 m²	Area $= \frac{90}{360} \times \frac{22}{7} \times 28^2$; Area $= 616$ m^2
8	**(b)** 120°	$\theta = \frac{462\times360}{\frac{22}{7}\times21^2}$; $\theta = 120^\circ$
9	**(c)** ₹ 2112	Cost $= 264 \times 8$; Cost = ₹ 2112
10	**(a)** 100	Distance 132 m $= 13200$ cm; revolutions $= \frac{13200}{132}$; Revolutions $= 100$

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Grazed area is a quadrant: $\frac{1}{4}\pi r^2$; Increase $= \frac{1}{4} \times \frac{22}{7}(21^2 - 7^2)$ (1); Increase $= 308$ m^2 (1).	1+1
12	Angle turned in 9 minutes $= 54^\circ$; Arc $= \frac{54}{360} \times 2 \times \frac{22}{7} \times 14$ (1); Distance $= 13.2$ cm (1).	1+1
13	Perimeter of the square = circumference $= 2\pi r = 264$ cm; Side $= 264 \div 4 = 66$ cm (1); Area $= 66^2 = 4356$ cm^2 (1).	1+1
14	Grazed area is a quadrant: $\frac{1}{4}\pi r^2$; Increase $= \frac{1}{4} \times \frac{22}{7}(28^2 - 14^2)$ (1); Increase $= 462$ m^2 (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Segment = sector − triangle (1); Sector $= 962.5$ cm^2, triangle $= \frac{1}{2} \times 35 \times 35 = 612.5$ cm^2 (1); Segment $= 962.5 - 612.5 = 350$ cm^2 (1).	1+1+1
16	Perimeter = arc $+ 2r$ (1); Arc $= \frac{135}{360} \times 2\pi \times 14 = 33$ cm (1); Perimeter $= 33 + 28 = 61$ cm (1).	1+1+1
17	The 4 ribs divide the circle into 4 equal sectors (1); Area of the circle $= \frac{22}{7} \times 14^2 = 616$ cm^2 (1); Area between ribs $= 616 \div 4 = 154$ cm^2 (1).	1+1+1
	OR: Segment = sector − triangle (1); Sector $= 1386$ cm^2, triangle $= \frac{1}{2} \times 42 \times 42 = 882$ cm^2 (1); Segment $= 1386 - 882 = 504$ cm^2 (1).	1+1+1
18	Perimeter = arc $+ 2r$ (1); Arc $= \frac{225}{360} \times 2\pi \times 14 = 55$ cm (1); Perimeter $= 55 + 28 = 83$ cm (1).	1+1+1
	OR: The 16 ribs divide the circle into 16 equal sectors (1); Area of the circle $= \frac{22}{7} \times 56^2 = 9856$ cm^2 (1); Area between ribs $= 9856 \div 16 = 616$ cm^2 (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Segment = sector − triangle (2); Sector $= 462.00$ cm^2; triangle with sides r, r and angle 120° has area $\frac{1}{2}r^2 \sin 120^\circ = \frac{\sqrt{3}}{4}r^2 = 190.73$ cm^2 (2); Segment $= 462.00 - 190.73 = 271.27$ cm^2 (1).	2+2+1
	OR: The diagonal of the square is the diameter $2r$, so its area is $\frac{(2r)^2}{2} = 2r^2$ (2); Circle $= 616$ cm^2; square $= 392$ cm^2 (2); Difference $= 224$ cm^2 (1).	2+2+1
20	Segment = sector − triangle (2); Sector $= 821.33$ cm^2; triangle with sides r, r and angle 120° has area $\frac{1}{2}r^2 \sin 120^\circ = \frac{\sqrt{3}}{4}r^2 = 339.08$ cm^2 (2); Segment $= 821.33 - 339.08 = 482.25$ cm^2 (1).	2+2+1
	OR: The diagonal of the square is the diameter $2r$, so its area is $\frac{(2r)^2}{2} = 2r^2$ (2); Circle $= 1386$ cm^2; square $= 882$ cm^2 (2); Difference $= 504$ cm^2 (1).	2+2+1

Mistakes to avoid:

- Using πr^2 in the arc-length formula or $2\pi r$ in the sector-area formula.
- Forgetting to add the two radii when asked for the *perimeter* of a sector.
- Using diameter in place of radius in πr^2.

- Subtracting the wrong figure for a segment: minor segment $=$ sector $-$ triangle, not sector $+$ triangle.
- Mixing units (cm and m) or writing cm instead of cm^2 for areas.
- Thinking the area of a circle doubles when the radius doubles; it becomes four times.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

MASTERMIND